Solidigm™ D7-PS1010 Gen5 NVMe

MLPerf Storage v2.0 Submission

System Name: Solidigm D7-PS1010 Gen5 NVMe

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Benchmark Version: v2.0

Solution Type: Local Storage

Submission Division: CLOSED

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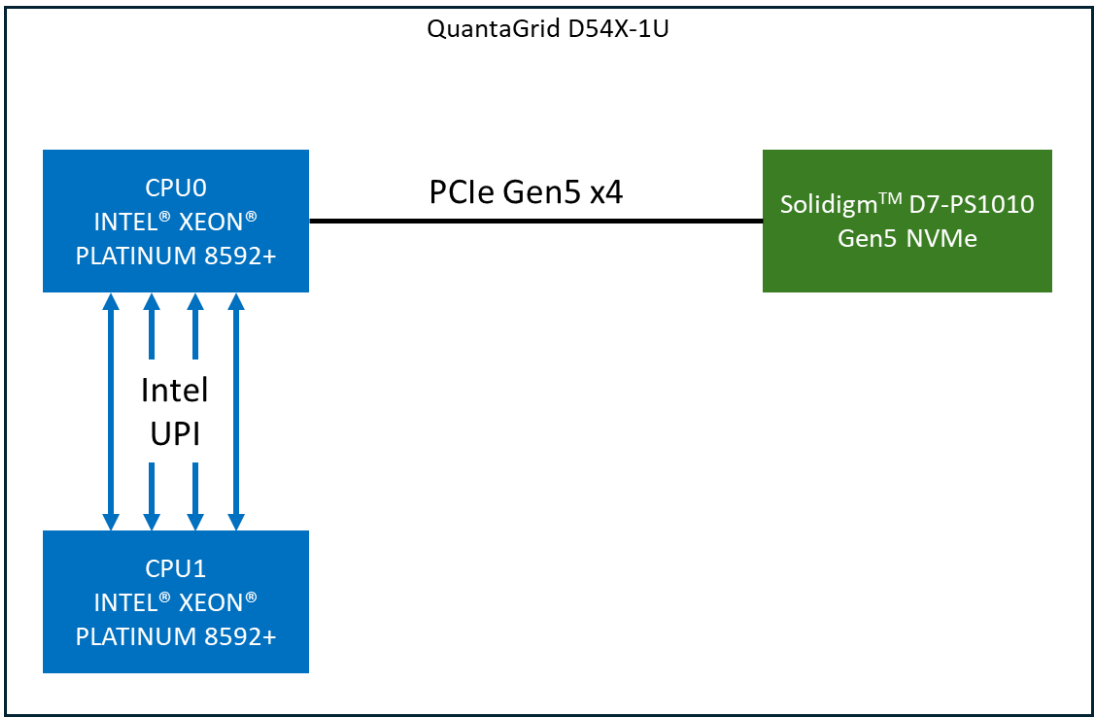
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1. Hardware

The hardware setup uses a 1U single server with locally installed NVMe drives. In the system topology, the NVMe is directly connected to CPU0 via a PCIe Gen5 x4 link.





Power Requirement Table (Direct-attached media)

System components	Power supply unit	Nameplate rated power	Design Power
Solidigm™ D7-PS1010 Gen5 NVMe	Active	23W	23W
Totals		**23W**	**23W**

2. Software

The distribution is Rocky Linux release 9.5 (Blue Onyx), with kernel version 5.14.0-503.14.1.el9_5.x86_64. The UCX library required by OpenMPI is provided by the built-in ucx-1.18.0 included in NVIDIA DOCA-OFED-2.9.2. The disk file system is formatted as XFS, using the default parameters for mkfs.xfs.

3. Settings

The system itself was not heavily customized, and only the following basic configurations were applied:

- **BIOS Settings**
  - Set the CPU profile to **High Performance**
  - **Disabled CPU SNC** (resulting in a single NUMA domain per CPU)
- **Operating System Settings**
  - Enabled tuned and set the profile to **throughput-performance**

```
sudo tuned-adm profile throughput-performance
```

  - Disabled kernel parameter "**numa_balancing**"

```
sudo sysctl -w kernel.numa_balancing=0
```